Omnichannel Weighted Scoring Model

The scoring process has 7 main calculations:

1. Success Rate
2. Frequency Score
3. Recency Score
4. Channel Engagement Score
5. Weighted Channel Contribution
6. Overall HCP Engagement Score
7. Engagement Level and Recommended Channel

1. Success Rate

For each HCP and each channel:

$$\boxed{Success\ Rate = \frac{Successful\ Interactions}{Total\ Interactions}}$$

Example — Email

Suppose HCP 101 has:

- Total Email interactions = 10
- Successful Email interactions = 6

Therefore:

$$Success\ Rate_{Email} = \frac{6}{10} = 0.60$$

So:

$$\boxed{Email\ Success\ Rate = 0.60 = 60\%}$$

2. Frequency Score

We need to normalize the number of interactions because different HCPs may have different numbers of interactions.

The formula used in our code is:

$$\boxed{Frequency\ Score = \frac{HCP's\ Interaction\ Count}{Maximum\ Interaction\ Count\ for\ that\ Channel}}$$

Example — Email

Suppose:

- HCP 101 has 10 Email interactions.
- The maximum Email interactions by any HCP = 20.

Therefore:

$$Frequency\ Score_{Email} = \frac{10}{20} = 0.50$$

So:

$$\boxed{Email\ Frequency\ Score = 0.50}$$

This means HCP 101 has achieved **50% of the highest observed Email interaction frequency**.

3. Recency Score

Recency tells us **how recently the HCP interacted through a particular channel.**

First:

$$Recency\ Days = Reference\ Date - Last\ Engagement\ Date$$

Then we calculate:

$$\boxed{Recency\ Score = \frac{1}{1 + \frac{Recency\ Days}{30}}}$$

The `30` represents our reference period of approximately one month.

Example

Suppose HCP 101's last Email interaction occurred **15 days ago**.

Therefore:

$$Recency\ Days = 15$$

Then:

$$Recency\ Score = \frac{1}{1+\frac{15}{30}}$$

$$= \frac{1}{1+0.5}$$

$$= \frac{1}{1.5}$$

$$\boxed{Recency\ Score = 0.667}$$

So the more recent the interaction, the higher the score.

For example:

Days Since Interaction	Recency Score
0	1.000
15	0.667
30	0.500
60	0.333
90	0.250
180	0.143

4. Channel Engagement Score

Now we combine the three components:

- Success
- Frequency
- Recency

We assigned:

Component	Weight
Success Rate	50%
Frequency Score	30%
Recency Score	20%

Therefore:

$$\boxed{Channel\ Score = 0.50(Success\ Rate) + 0.30(Frequency\ Score) + 0.20(Recency\ Score)}$$

Example — Email

We already calculated:

$$Success\ Rate = 0.60$$

$$Frequency\ Score = 0.50$$

$$Recency\ Score = 0.667$$

Therefore:

$$ChannelScore_{Email} = (0.50)(0.60) + (0.30)(0.50) + (0.20)(0.667)$$

$$= 0.30 + 0.15 + 0.1334$$

$$\boxed{Email\ Channel\ Score = 0.5834}$$

So the Email channel score is approximately:

$$\boxed{58.34/100}$$

5. Calculate Channel Scores for All Channels

Let's assume the following data for HCP 101:

Channel	Interactions	Successful	Maximum Interactions	Days Since Last Interaction
Email	10	6	20	15
Phone Call	5	2	10	30
Rep Visit	4	3	8	10
Webinar	3	1	6	60
Digital Ad	8	1	16	90

Now calculate each channel.

Email

$$Success = \frac{6}{10} = 0.60$$

$$Frequency = \frac{10}{20} = 0.50$$

$$Recency = \frac{1}{1 + 15/30} = 0.667$$

Therefore:

$$EmailScore = (0.50)(0.60) + (0.30)(0.50) + (0.20)(0.667)$$

$$\boxed{EmailScore = 0.583}$$

Phone Call

$$Success = \frac{2}{5} = 0.40$$

$$Frequency = \frac{5}{10} = 0.50$$

$$Recency = \frac{1}{1 + 30/30} = 0.50$$

Therefore:

$$PhoneScore = (0.50)(0.40) + (0.30)(0.50) + (0.20)(0.50)$$

$$= 0.20 + 0.15 + 0.10$$

$$\boxed{PhoneScore = 0.450}$$

Rep Visit

$$Success = \frac{3}{4} = 0.75$$

$$Frequency = \frac{4}{8} = 0.50$$

$$Recency = \frac{1}{1 + 10/30} = 0.75$$

Therefore:

$$RepVisitScore = (0.50)(0.75) + (0.30)(0.50) + (0.20)(0.75)$$

$$= 0.375 + 0.15 + 0.15$$

$$\boxed{RepVisitScore = 0.675}$$

Webinar

$$Success = \frac{1}{3} = 0.333$$

$$Frequency = \frac{3}{6} = 0.50$$

$$Recency = \frac{1}{1 + 60/30} = 0.333$$

Therefore:

$$WebinarScore = (0.50)(0.333) + (0.30)(0.50) + (0.20)(0.333)$$

$$\boxed{WebinarScore \approx 0.383}$$

Digital Ad

$$Success = \frac{1}{8} = 0.125$$

$$Frequency = \frac{8}{16} = 0.50$$

$$Recency = \frac{1}{1 + 90/30} = 0.25$$

Therefore:

$$DigitalAdScore = (0.50)(0.125) + (0.30)(0.50) + (0.20)(0.25)$$

$$= 0.0625 + 0.15 + 0.05$$

$$\boxed{DigitalAdScore = 0.263}$$

6. Final Channel Scores

So HCP 101 has:

Channel	Success Rate	Frequency Score	Recency Score	Channel Score
Email	0.600	0.500	0.667	**0.583**
Phone Call	0.400	0.500	0.500	**0.450**
Rep Visit	0.750	0.500	0.750	**0.675**
Webinar	0.333	0.500	0.333	**0.383**
Digital Ad	0.125	0.500	0.250	**0.263**

Notice that:

$$RepVisitScore = 0.675$$

is the highest.

This tells us that **Rep Visit is currently the strongest engagement channel for HCP 101.**

7. Assign the Channel Weights

Our current business-defined weights are:

Channel	Weight
Rep Visit	30%
Phone Call	20%
Webinar	20%
Email	15%
Digital Ad	15%
Total	**100%**

Mathematically:

$$0.30 + 0.20 + 0.20 + 0.15 + 0.15 = 1$$

The weights therefore satisfy:

$$\boxed{\sum W_c = 1}$$

8. Weighted Channel Contribution

Before calculating the final score, we calculate each channel's contribution.

Formula

$$\boxed{Weighted\,Contribution_c = ChannelScore_c \times ChannelWeight_c}$$

For HCP 101:

Rep Visit

$$0.675 \times 0.30 = \boxed{0.2025}$$

Phone Call

$$0.450 \times 0.20 = \boxed{0.0900}$$

Webinar

$$0.383 \times 0.20 = \boxed{0.0766}$$

Email

$$0.583 \times 0.15 = \boxed{0.0875}$$

Digital Ad

$$0.263 \times 0.15 = \boxed{0.0395}$$

9. Overall HCP Engagement Score

Now add all the weighted contributions.

Formula

$$\boxed{Overall\ Score = \sum(ChannelScore \times ChannelWeight)}$$

For HCP 101:

$$OverallScore = (0.675)(0.30) + (0.450)(0.20) + (0.383)(0.20) + (0.583)(0.15) + (0.263)(0.15$$

$$= 0.2025 + 0.0900 + 0.0766 + 0.0875 + 0.0395$$

$$\boxed{OverallScore \approx 0.496}$$

10. Convert to a 0–100 Score

For easier interpretation:

$$\boxed{Final\ Engagement\ Score = OverallScore \times 100}$$

Therefore:

$$FinalScore = 0.496 \times 100$$

$$\boxed{Final\ Engagement\ Score \approx 49.60}$$

So HCP 101 has:

> **Engagement Score = 49.60 / 100**

11. Engagement Level

We then classify the HCP according to the final score.

Our current thresholds are:

Score	Engagement Level
0–20	Low
>20–40	Medium
>40–100	High

HCP 101 has:

$$49.60$$

Since:

$$49.60 > 40$$

we classify:

$$\boxed{High\ Engagement}$$

12. Engagement Rank

After calculating the final score for **all 500 HCPs**, we rank them from highest to lowest.

Formula

$$\boxed{Rank_h = RankDescending(FinalScore_h)}$$

For example:

HCP	Score	Rank
HCP 201	82.40	1
HCP 145	76.30	2
HCP 101	49.60	3
HCP 318	41.20	4
HCP 412	18.70	5

In this hypothetical example, HCP 101 would have:

$$\boxed{Rank = 3}$$

The actual rank will depend on the scores of all 500 HCPs.

13. Recommended Channel

The basic recommendation is based on the highest channel score.

Formula

$$\boxed{RecommendedChannel = \arg\max(ChannelScores)}$$

For HCP 101:

Channel	Score
Email	0.583
Phone Call	0.450
Rep Visit	**0.675**
Webinar	0.383
Digital Ad	0.263

The highest score is Rep Visit.

Therefore:

$$\boxed{RecommendedChannel = RepVisit}$$

14. Opt-Out Constraint

The `opt_out_flag` should be treated separately from the engagement score.

Suppose:

$$FinalEngagementScore = 73$$

but:

$$opt\_out\_flag = True$$

We should **not change the historical score to zero**.

The score means:

> "This is how engaged the HCP has historically been."

The opt-out flag means:

> "The HCP should not be contacted."

Therefore:

$$\boxed{Historical\ Score \neq Contact\ Eligibility}$$

The final recommendation becomes:

$$\boxed{Do\ Not\ Contact}$$

rather than Email/Phone/Rep Visit/etc.

Final interpretation for HCP 101

Putting everything together:

$$\boxed{EmailScore = 0.583}$$

$$\boxed{PhoneScore = 0.450}$$

$$\boxed{RepVisitScore = 0.675}$$

$$\boxed{WebinarScore = 0.383}$$

$$\boxed{DigitalAdScore = 0.263}$$

Then:

$$\boxed{OverallScore \approx 0.496}$$

and:

$$\boxed{FinalEngagementScore \approx 49.60/100}$$

Therefore:

HCP 101 → 49.60 Engagement Score → High Engagement → Rep Visit as the recommended channel.

Therefore:

HCP 101 → 49.60 Engagement Score → High Engagement → Rep Visit as the recommended channel.

<u>Note:</u>

The **50%-30%-20% component weights, 30%-20%-20%-15%-15% channel weights and 20/40 engagement thresholds are the current model assumptions**. We should document them as business-rule parameters and later perform a **sensitivity/validation analysis** to justify whether these values are optimal.

******The End******